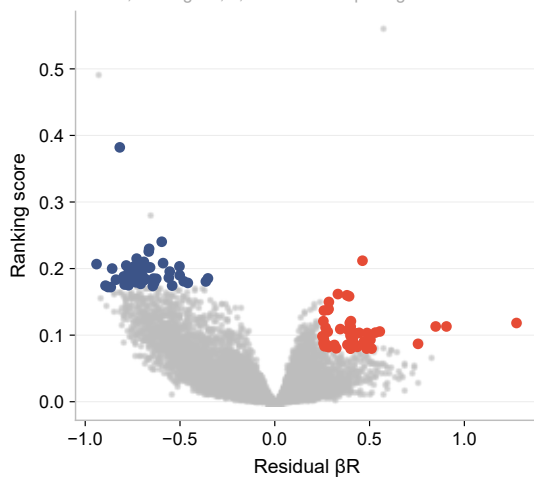


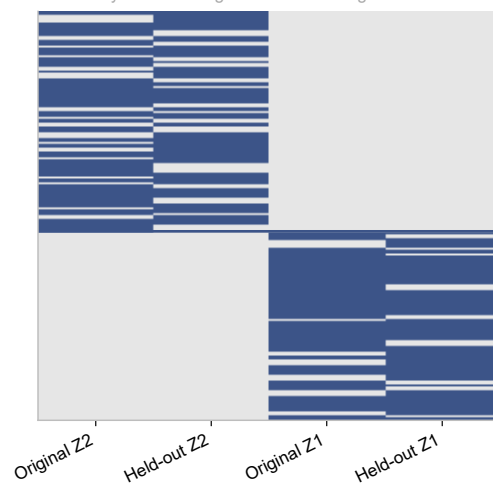
A Frozen 100-gene selection

24,356 eligible; 3,560 threshold-pass genes



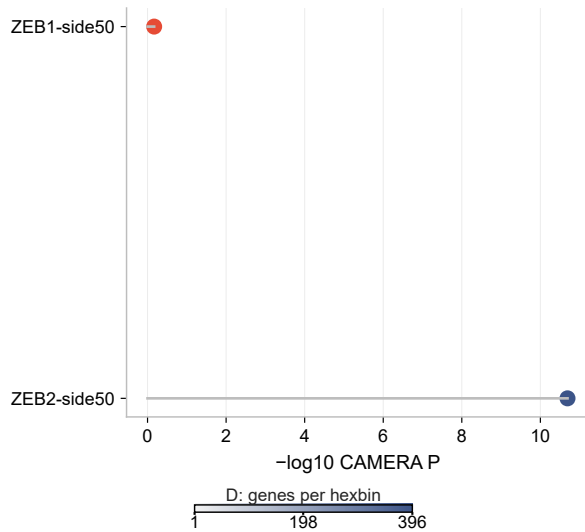
B Frozen membership overlap

Binary matrix of original and held-out gene sets



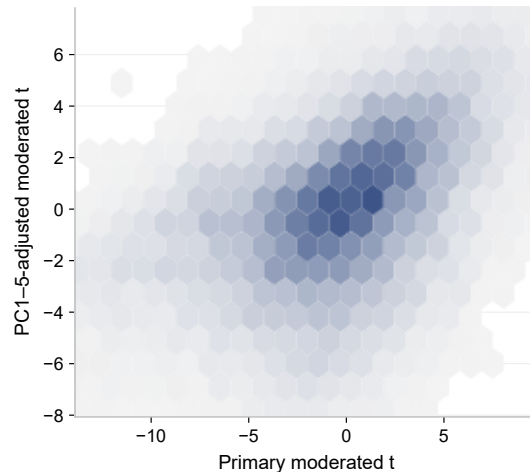
C Held-out CAMERA in matched pairs

Within-cohort phenotype bridge



D PC-adjustment sensitivity

Key-aligned gene statistics; descriptive



B: frozen-set membership

■ present

■ absent